BANK CUSTOMER CHURN ANALYSIS PROJECT

Tools Used: SQL | Power BI
Dataset Source: Kaggle
Dataset Format: CSV

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1. Business Problem

Banks are losing customers because some of them close their accounts and leave, which is known as **customer churn**. High customer churn directly impacts the bank's revenue, growth, and long-term stability.

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2. Business Goal

The goal of this project is to identify customer groups with higher churn and analyze the key factors associated with customer churn.

The project aims to:

- Identify customer groups with higher churn.
- Analyze key factors associated with churn, such as **Age, Balance, Credit Score, Geography, and Active Member Status**.
- Help the bank develop customer retention strategies to reduce churn and improve customer loyalty.

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3. Business Questions

1. What is the overall customer churn rate?
2. Which age group has the highest customer churn?
3. Which balance group has the highest customer churn?
4. Is customer churn higher in certain geographies, such as France, Spain, or Germany?
5. Do inactive members churn more than active members?

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4. Dataset and Tools

Dataset Details

- **Dataset Name:** Bank Customer Churn Dataset
- **Data Source:** Kaggle
- **File Format:** CSV
- **Total Customers:** 10,000

Tools Used

SQL

SQL was used for:

- Data cleaning
- Data validation
- Data analysis

Power BI

Power BI was used for:

- Creating an interactive dashboard
- Creating KPI cards
- Visualizing customer churn patterns
- Analyzing churn by age group, geography, balance group, and member status

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5. Project Workflow

Kaggle Dataset → CSV File → SQL Data Cleaning → SQL Data Validation → SQL Business Analysis → Data Export → Power BI → Interactive Dashboard

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6. SQL Data Cleaning

The Bank Customer Churn dataset was imported into SQL for data cleaning and preparation.

The dataset was checked for:

- Total row count
- Data types
- Missing or null values
- Duplicate Customer IDs
- Special characters in customer names
- Geography and Gender values
- Invalid values
- Outliers using minimum and maximum values
- Extra spaces in Geography and Gender columns

The incorrectly encoded **Row Number** column name was corrected to improve data readability and consistency.

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7. SQL Data Validation

After the data cleaning process, the dataset was revalidated to confirm that it was suitable for analysis.

The following checks were performed:

- Total row count was rechecked.
- Invalid values were rechecked.
- Data consistency was verified.

The validation process helped ensure that the dataset was ready for business analysis and visualization.

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8. SQL Analysis and Key Insights

1. Overall Customer Churn Rate

Result:

- Total Customers: **10,000**

- Churn Customers: **2,037**
- Overall Churn Rate: **20.37%**

Insight:

Approximately one out of every five customers has left the bank. This indicates that customer churn is a significant business concern and requires focused retention strategies.

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2. Customer Churn by Age Group

Result:

- Middle-aged: **1,350**
- Senior: **563**
- Young: **124**

Insight:

Middle-aged customers have the highest number of churn customers. The bank should further analyze the needs and experience of this customer group to reduce churn.

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3. Customer Churn by Balance Group

Result:

- High Balance: **1,211**
- Zero Balance: **500**
- Medium Balance: **300**
- Low Balance: **26**

Insight:

The High Balance group has the highest number of churn customers. The bank should focus on retaining these customers because they may represent significant financial value.

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4. Customer Churn by Geography

Result:

- Germany: **814**
- France: **810**
- Spain: **413**

Insight:

Germany has the highest number of churn customers, closely followed by France. The bank should investigate regional factors that may be contributing to higher churn.

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5. Customer Churn by Member Status

Result:

- Inactive Members: **1,302**
- Active Members: **735**

Insight:

Inactive members have a higher number of churn customers than active members. Increasing customer engagement may help reduce churn.

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9. Power BI Dashboard

After completing data cleaning, data validation, and business analysis in SQL, the analyzed data was imported into Power BI.

An interactive **Bank Customer Churn Analysis Dashboard** was created to present the key findings in a clear and user-friendly format.

The dashboard includes:

- Total Customer KPI
- Churn Customer KPI
- Churn Rate KPI
- Average Credit Score KPI
- Overall Customer Churn Rate Donut Chart
- Customers Churn by Age Group

- Churn Customers by Geography
- Churn Customer by Balance Group
- Churn Customers by Member Status
- Gender Filter
- Age Filter
- Geography Filter

The dashboard allows users to interactively filter and analyze customer churn based on gender, age group, and geography.

Dashboard Screenshot

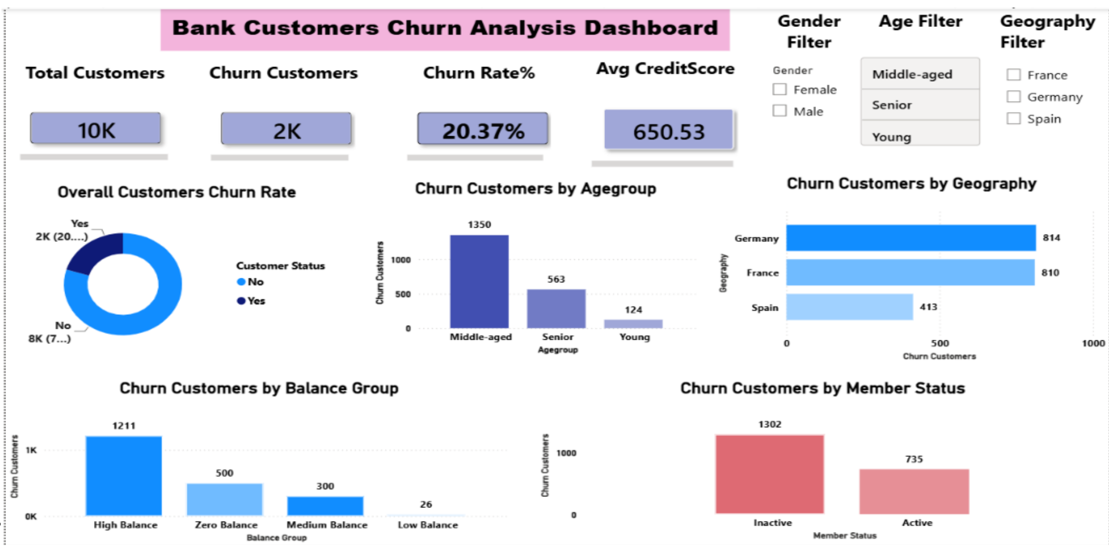

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Figure 1: Interactive Bank Customer Churn Analysis Dashboard created using Power BI

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10. Recommendations

1. Focus more on middle-aged customers because they have the highest churn.
2. Provide special offers and better services to high-balance customers to encourage them to stay with the bank.
3. Check the reasons for high churn in Germany and France and improve customer services in these regions.

4. Encourage inactive customers to use bank services through reminders, offers, and regular communication.
5. Use the Power BI dashboard to regularly monitor customer churn and identify customer groups that need more attention.

11. Conclusion

This project analyzed bank customer churn data using SQL and Power BI. SQL was used for data cleaning, validation, and analysis, while Power BI was used to create an interactive dashboard.

The analysis identified key churn patterns across age groups, balance groups, geographies, and member status. The findings can help the bank identify high-risk customer groups, improve customer engagement, and develop effective retention strategies to reduce customer churn.